To count or to predict? Revisiting error-driven distributional learning of phonetic categories

Erdin Mujezinović & Ruben van de Vijver

erdin.mujezinovic@hhu.de, ruben.vijver@hhu.de

Department of General Linguistics
Heinrich-Heine University Düsseldorf, Germany

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To count or to predict?

Revisiting error-driven distributional learning of phonetic categories

How do native speakers acquire their language specific phonetic categories?

- Certain phonetic dimensions are used to differentiate meaning
- The same phonetic category can vary in its phonetic implementation cross-linguistically
- Understanding the learning mechanisms = understanding how and why these categories (and differences) arise

To count or to predict?

Revisiting error-driven distributional

Two conceptualizations:

Distributional learning

Contiguity: Counting co-occurrences (Maye, Werker, & Gerken, 2002; Saffran, 2020; Saffran, Aslin, & Newport, 1996; Saffran & Kirkham, 2018).

Error-driven learning

Contingency: Using cues to predict outcomes (Hoppe et al., 2022; Nixon, 2020; Nixon & Tomaschek, 2021; Ramscar, Dye, & McCauley, 2013; Rescorla & Wagner, 1972).

Distributional learning

Maye, Werker, & Gerken (2002)

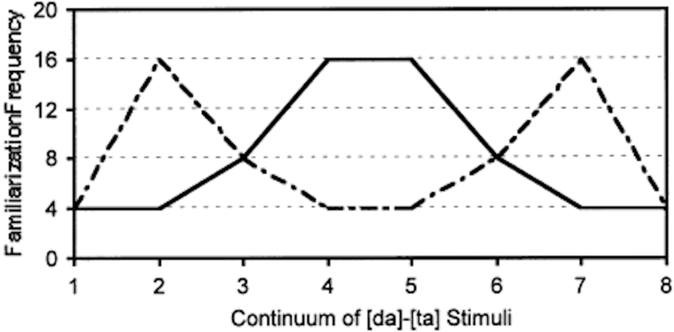

Figure 1: Unimodal (drawn line) vs. bimodal (dashed line) distribution (Maye, Werker, & Gerken, 2002)

Distributional learning

Maye, Werker, & Gerken (2002)

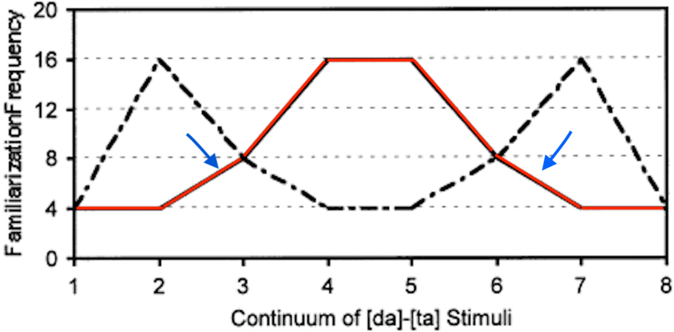

Figure 2: Unimodal (drawn line) vs. bimodal (dashed line) distribution (Maye, Werker, & Gerken, 2002)

- What if the tails are not symmetrical (not normally distributed?)

Error-driven distributional learning

Olejarczuk, Kapatsinski, & Baayen (2018)

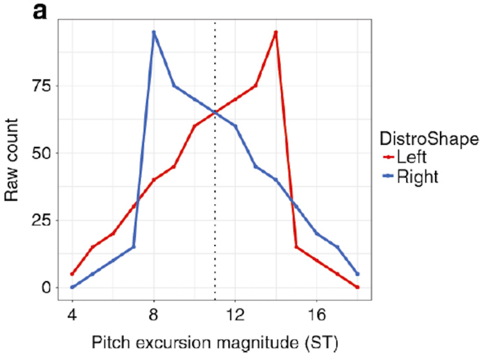

Figure 3: Distribution of LHL syllables ([ka]) for both groups in Olejarczuk, Kapatsinski, & Baayen (2018)

- Original result: Rarer tokens have a disproportionate influence on the category

Our study

The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
- e.g. Japanese [oto] “sound” vs. [otːo] “husband”
- Original effect found for the learning of a rise-fall tonal category with English speakers
  - Problem: Pitch is not used at the lexical stratum in English
  - Effect found in Olejarczuk, Kapatsinski, & Baayen (2018) may be limited to non-lexical categories (such as tone)
  - German uses duration lexically for vowels (but not consonants)
  - /a/ vs. /aː/ in *Bahn* [baːn] vs. *Bann* [ban]

Our study

What phonetic category had to be learned?

Gemination

Segment duration of lateral approximant [lː] in [alːa]-sequence manipulated in 20 ms increments using **Praat** (Boersma & Weenink, 2021)

Our study

The stimuli

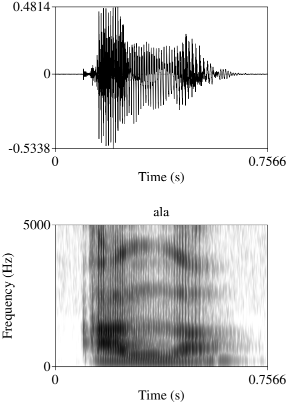

Figure 4: The original [alːa]-stimulus

Our study

The stimuli

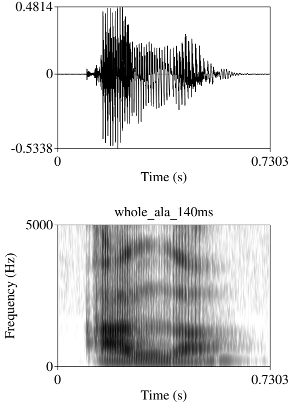

Figure 5: Shorter [alːa]-stimulus (140 ms)

Our study

The stimuli

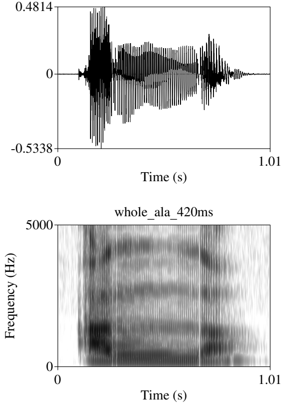

Figure 6: Longer [alːa]-stimulus (420 ms)

Our study

The experimental groups

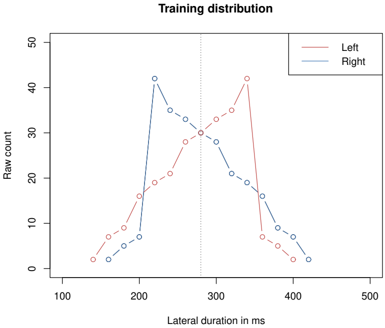

Figure 7: Right skew (blue) and left skew (red)

1. Participants in both groups had to judge the [alːa] tokens for their *Category Goodness* using a 7-point Likert-scale
2. Participants had to produce three representative [alːa]-tokens

Our study

The experimental groups

Group	Range	Modal	Mean	Tokens
Right-skew	160-420 ms	220 ms	280 ms	256
Left-skew	140-400 ms	340 ms	280 ms	256
Test	80-480 ms			21

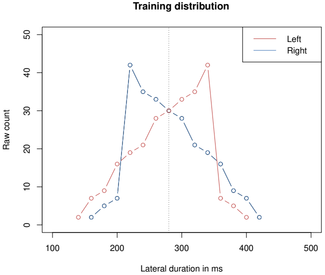

Figure 8: Distribution of the different segment durations

Our study

Research question

Research question

Do less frequent tokens have a stronger influence on the phonetic category representation (in this case /lː/)?

Our study

Hypotheses

Three hypotheses:

1. Representation is based on the whole distribution only **(the mean)**
   - **No difference** between groups expected
2. Representation is additionally influenced by the most frequent tokens **(the modal values)**
   - For Left skew = **higher durations** are more representative
3. Representation is additionally influenced by the rarer tokens **(the skew)**
   - For Left skew = **lower durations** are more representative

The former two are in line with distributional learning (count), the latter one with error-driven learning (predict)

Our study

Results I: Category-Goodness

Overall: 294 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 189 observations (9 participants x 21 stimuli)
- Right-skew: 105 observations (5 participants x 21 Stimuli)
- Dataanalysis done in R (R Core Team, 2021)

Our study

Results I: Category-Goodness

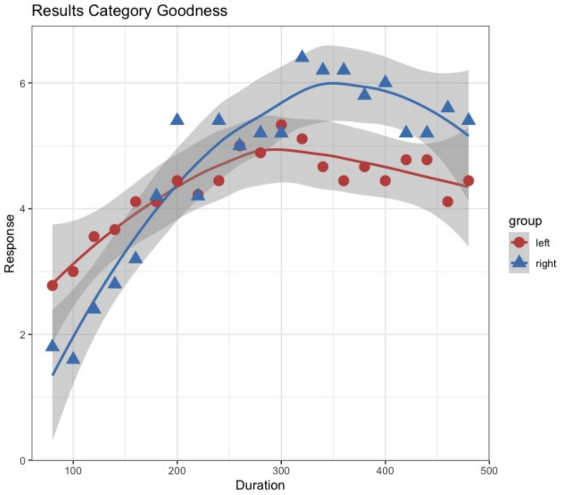

Figure 9: Overall results 7-point Likert scale (grey = Standard Error)

Our study

Results I: Category-Goodness

```
gam(response ~ group + s(stimulus, by = subject),
data = daten
```

Table 1: Summary table GAM Model (mgcv-package (Wood, 2011)).

A.Parametric coefficients					
	Estimate	Std. Error	t-value	Pr(>|t|)	
(Intercept)	4.1846	0.2814	14.87	< 0.0001	***
B.Smooth terms					
	edf	Ref.df	F	p-value	
s(stimulus):groupleft	1.996	2.230	2.549	0.08985	.
s(stimulus):groupright	2.161	2.429	4.816	0.00762	**
s(stimulus,subject)	62.890	125.000	3.020	< 0.0001	***

Our study

Results II: Production

Overall: 42 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 27 observations (9 participants x 3 stimuli)
- Right-skew: 15 observations (5 participants x 3 stimuli)
- Dataanalysis in R (R Core Team, 2021)

Our study

Results II: Production

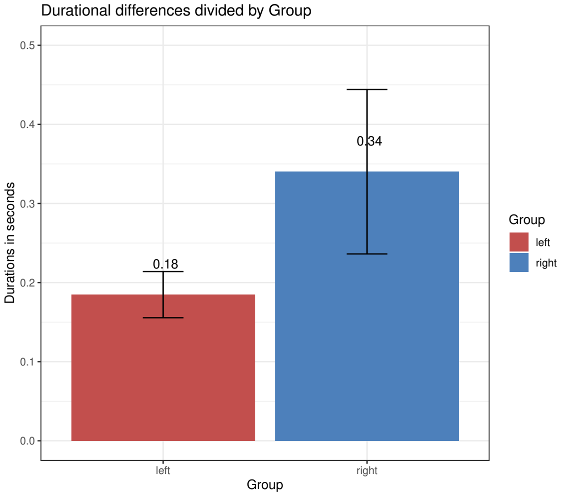

Figure 10: Difference in consonant duration between groups

Our study

Results II: Production

Linear mixed effects model computed using `lmerTest`-package (Kuznetsova, Brockhoff, & Christensen, 2017):

- Random intercepts for subject and item
- Significant effect for group (Est. = 0.1575, SE = 0.0630, df = 12.0041, = t = 2.501, p = 0.02785)
- Consonant duration differed between groups!

Discussion

Comparison to Olejarczuk, Kapatsinski, & Baayen (2018)

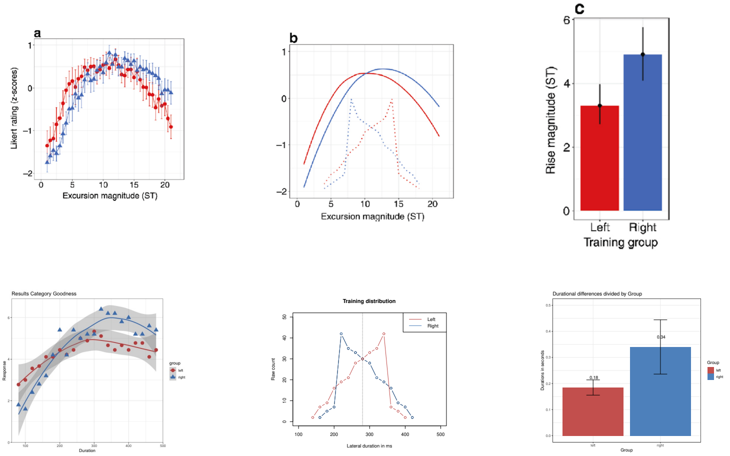

Figure 11: Comparison original study vs. replication

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

- Less frequent tokens have a stronger influence on the phonetic category ✓
- Learners do not simply count co-occurrences, but they instead **predict** subsequent tokens (outcomes) by using previous tokens as cues
  - Error-driven learning predicts such an effect
  - Rare category tokens are more surprising, generate more error and therefore cause stronger adjustment of the phonetic category
  - Frequent tokens define the category, but rarer instances constrain it!

Conclusion

Thank you for your attention!

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